

Musa Ibne Mannan

Dallas, Texas 75252 — (737) 221-1884 — musa314@utdallas.edu

Professional Summary

Ph.D. Candidate in Mechanical Engineering with extensive hands-on experience independently designing, building, troubleshooting, and operating complex mechanical systems and laboratory equipment. Strong foundation in solid mechanics, dynamics, and finite element analysis (Abaqus) applied to mechanical component performance and failure analysis. Skilled in Python and MATLAB for data analysis, CAD (SolidWorks, AutoCAD, Catia), and rapid prototyping/fabrication. Demonstrated ability to design and execute custom experiments, communicate technical findings clearly, and deliver rigorous results under deadlines. Eager to apply experimental, analytical, and investigative skills to forensic mechanical engineering, failure analysis, and technical consulting.

Education

Doctor of Philosophy (Ph.D.) in Mechanical Engineering Fall 2023 – Present

Erik Jonsson School of Engineering and Computer Science, The University of Texas at Dallas

Concentration: Manufacturing and Design Innovations CGPA: 3.62

Master of Science in Mechanical & Manufacturing Engineering (ABET) Aug 2021 – May 2023

Ingram School of Engineering, Texas State University, San Marcos, TX CGPA: 3.75

Professional Experience

Research Assistant & Lab In-Charge Jun 2024 – Present

Leem Research Group, University of Texas at Dallas, TX

- Independently designed, assembled from the ground up, troubleshoot, and operated high-vacuum CVD furnaces, hydrogen generator, and integrated Raman spectroscopy instrumentation; and performed process optimization.
- Led complete laboratory build-out including procurement, equipment installation, safety compliance, and inventory management while executing concurrent computational and experimental research programs.
- Applied Python-based data analysis pipelines and ensemble machine learning (Random Forest, XGBoost) to optimize multi-parameter mechanical system performance and materials synthesis processes.

Graduate Student Teaching Associate (Primary Instructor) Aug 2025 – Present

University of Texas at Dallas, TX

- Serve as sole instructor for ECS 1100, responsible for curriculum delivery, mentoring, and assessment of 130 first-year engineering students; develop and deliver clear technical lectures on engineering fundamentals including mechanics and design.
- Demonstrate strong ability to distill and communicate complex technical concepts to diverse audiences through lectures, instructional materials, and one-on-one support.

Assistant / Site Engineer Sep 2018 – Jun 2019

Washillah Engineering Services (Partex Petro Ltd. & CUFL projects), Bangladesh

- Led on-site teams of 28–35 engineers and contractors for large-scale mechanical and piping projects (condensate refinery and boiler re-tubing); enforced EHS/OSHA standards and managed field operations under tight deadlines.
- Performed root-cause analysis and resolved over 120 design and implementation errors in mechanical systems and structures, saving approximately 30,000 man-hours and \$1M while maintaining quality and schedule.
- Coordinated cross-functional teams, client interfaces, material logistics, and quality control in high-stakes industrial field environments.

Graduate Instructional Assistant Aug 2023 – May 2024

University of Texas at Dallas, TX

- Graded assignments/exams and tutored 140+ students in Kinematics & Dynamics (MECH 3350) and Mechanics of Materials (MECH 2320)
- Delivered substitute lectures and assisted students with computational modeling, FEA, and CAD troubleshooting.

Texas State University, TX

- Mentored three senior design teams (15 students) to successful completion on industry-sponsored mechanical engineering capstone projects; provided hands-on guidance in CAD, FEA, prototyping, and mechanical system design.

Technical Skills

Hands-on Experimental, Lab & Fabrication: Independent design and execution of custom experiments, CVD system assembly/troubleshooting, Raman spectroscopy and materials characterization (SEM/EDX/XPS/XRD/TGA/DSC), FDM/SLA/Markforged 3D printing, laser cutting, waterjet, TIG/Arc welding, manual mill/lathe, shop fabrication and rapid prototyping.

Computational & Analysis: Python (data analysis pipelines, Random Forest, XGBoost), MATLAB (numerical methods, optimization), Abaqus (transient heat transfer, stress analysis, mesh convergence, mechanical component FEA), SolidWorks, AutoCAD, Catia, basic CFD.

Technical Communication & Documentation: Technical report writing, publication-quality figures and presentations, high-accuracy documentation under time pressure, Adobe Creative Suite.

Publications

1. Wasi Shadman, Musa Mannan, et al. "Electrochemical detection of tRNA-derived fragments (tRFs) based on multi-functionalized carbon surface." *Surfaces and Interfaces* (Under Review).
2. MSHK Tushar, MI Mannan, A Azmain. "Exhaust Gas After Treatment using Air Preheating and Selective Catalytic Reduction by Urea to Reduce NOx in Diesel Engine." *Heliyon*, 11(1), 2025.
3. MI Mannan. "In-Vitro Detection of tRNA Fragments (tRFs) Using an Inkjet-Printed Graphene Electrochemical Aptasensor." M.S. Thesis, Texas State University, 2023.
4. S Panta, MI Mannan, et al. "Comparison of Mechanical and Morphological Properties of Amino and Glymo Silane Treated Ragweed-PLA Bio-Composites." *CAMX 2022 Proceedings*, Anaheim, CA, 2022.
5. KM Jitendra Tate, Musa Ibne Mannan, et al. "Thermal and Structural Properties of Surface Treated Ragweed Fiber Reinforced Biodegradable Polylactic Acid Biocomposites." *International Journal of Composite and Constituent Materials*, 8(2), 2022.

Selected Projects — Mechanical Systems Analysis & Failure Characterization

- **Machine Gun Barrel Thermal & Stress Failure Analysis (Abaqus FEA):** Performed transient heat transfer and von Mises stress analysis on barrel materials under sustained high-rate firing; evaluated AISI 4140, 316L, and ASTM A656 for thermal gradients, stress concentrations, and mechanical failure risk
- **Biocomposite Mechanical System Characterization & Failure Diagnosis:** Designed, extruded, and 3D-printed silane-treated natural fiber composites; used SEM/EDAX, tensile/flexural testing, and thermal analysis to diagnose interfacial adhesion issues, fiber-size effects, and resulting mechanical failure modes.
- **rGO Thin-Film Processing & Comprehensive Characterization:** Independently designed and executed Pulsed Laser Deposition + Annealing experiments; performed full materials characterization (XPS, Raman, XRD, electrical)

Leadership & Additional Professional Experience

Emergency Room Scribe Workflow Innovation, Augmedix

Mar 2020 – Jul 2021

Designed and piloted the world's first remote ER medical scribing program; created training curriculum from scratch and implemented new high-stakes real-time documentation workflow across multi-site teams (featured in Business Insider).

Medical Scribe, Augmedix

Jul 2019 – Jul 2021

Delivered real-time documentation support for 4,000+ patients across multiple medical disciplines; demonstrated rapid technical domain learning, meticulous attention to detail, and performance under timed, high-pressure conditions.

Novelist (Bangla)

Feb 2016 – Present

Author of 17 published crime novels; developed strong skills in clear technical narrative, accuracy, and effective communication.